



**G H RAISONI COLLEGE OF ENGINEERING AND
MANAGEMENT, PUNE**

**AN EMPOWERED AUTONOMOUS INSTITUTE, AFFILIATED
TO SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE**

Submitted for

TAE-I

in the subject

Database Management System (DBMS)

Submitted By:

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**G H RAISONI COLLEGE OF ENGINEERING AND
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1.Title:

Futures & Options Trading, Volatility and Settlement Management Database System

2. About the Database System

The Futures & Options Trading, Volatility and Settlement Management Database System is a relational database that integrates the F&O Volatility and Derivatives Bhavcopy datasets. It stores security, trading, price, settlement, and volatility information in normalized tables using primary and foreign keys. The system ensures efficient data management, reduces redundancy, and supports SQL operations such as joins, queries, reports, and analysis.

Dataset 1: [F&O Volatility.csv](#)

Dataset 2: [F&O Volatility.csv](#)

3. Problem Statement & System Requirements

3.1 Problem Statement

The Futures & Options Trading, Volatility and Settlement Management Database System is developed to integrate and manage data from the F&O Volatility Dataset and the Derivatives Bhavcopy Dataset. The system stores information related to securities, trading dates, financial instruments, market prices, settlement prices, expiry dates, and volatility measures in a structured relational database.

The project aims to eliminate data redundancy, maintain data integrity, and support efficient storage, retrieval, and analysis of trading and volatility information using relational database concepts.

3.2 Business Domain

Domain: Financial Market – Futures & Options (F&O) Trading

The database focuses on managing daily trading records, futures contracts, settlement prices, and volatility information for securities traded in the derivatives market.

3.3 Entities:

1. Security
 2. Market Record
 3. Trading Date
 4. Price Details
 5. Instrument
 6. Volatility
 7. Expiry
 8. Risk Assessment
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4. Literature & System Reference

4.1 Introduction

The proposed Futures & Options Trading, Volatility and Settlement Management Database System is designed using official NSE datasets, relational database concepts, and standard database design methodologies. The database integrates trading, settlement, and volatility information into a normalized relational schema to support efficient storage, querying, and analysis.

4.2 Research Paper 1

Title

Volatility Forecasting Using Historical Stock Market Data

Objective

To analyze historical stock and futures price movements for predicting market volatility.

Contribution:

- Explains volatility measurement techniques.
- Uses historical price and return data.
- Helps in designing the **Volatility** entity.

4.3 Research Paper 2

Title

Database Design for Financial Trading Systems

Objective

To develop an efficient relational database for financial transactions.

Contribution

- Entity Relationship Modeling
- Normalization
- Data Integrity
- Primary and Foreign Keys

4.4 Technical Documentation

National Stock Exchange (NSE)

Datasets Used

- F&O Volatility Dataset

Information Obtained

- Symbol
- Instrument
- Expiry Date
- Open Price
- High Price
- Low Price
- Close Price
- Settlement Price
- Open Interest
- Trading Volume
- Volatility

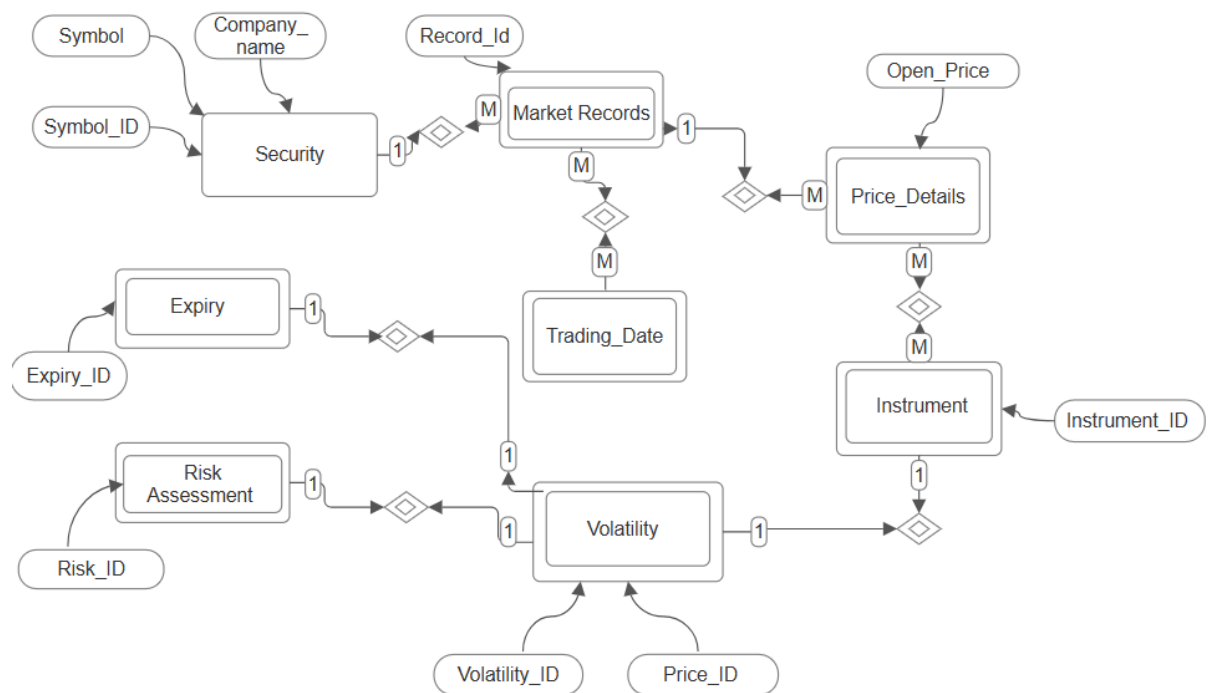
Contribution

Provides the real-world data used for database implementation.

4.5 Literature Summary

The reviewed literature and official NSE datasets provide the theoretical and practical foundation for designing a normalized relational database capable of storing trading, settlement, and volatility information while maintaining consistency and integrity.

5.ER Diagram



6. Relational Schema & Normalisation Plan

6.1 Relational Schema

- The ER diagram is converted into relational tables.
- The main tables are Security, Trading_Date, Instrument, Market_Record, Price_Details, Volatility, Expiry, and Risk_Assessment.
- Each table contains a Primary Key (PK) for unique identification.
- Foreign Keys (FK) are used to establish relationships between tables.
- Data types and integrity constraints are assigned to ensure data consistency.

6.2 Normalisation Plan

- **UNF:** Raw datasets contain repeated and redundant information.
- **1NF:** Data is divided into atomic values, and repeating groups are removed.
- **2NF:** Partial dependencies are eliminated by separating data into related tables.
- **3NF:** Transitive dependencies are removed to reduce redundancy.
- **BCNF:** Every determinant is a candidate key, ensuring a well-structured database.

6.3 Benefits

- Reduces data redundancy.
 - Improves data integrity and consistency.
 - Enhances query performance.
 - Simplifies database maintenance and updates.
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7. Target RDBMS Software

7.1 Selected RDBMS

- **MySQL**

7.2 Reasons for Selection

- Open-source and widely used relational database.
 - Supports **Primary Key, Foreign Key, and integrity constraints**.
 - Provides SQL features such as **joins, views, triggers, stored procedures, and transactions**.
 - Allows easy import of CSV datasets.
 - Efficiently manages large financial trading datasets.
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8.References:

- <https://www.nseindia.com/all-reports-derivatives> (NSE Datasets)
- <https://app.smartdraw.com/editor.aspx?templateId=d6ac67f7-8fe8-4e0a-bcff-b88d36345596&flags=128#credID=-100017608&depoId=76243785> (ER Diagram Editor)
- Oracle Corporation, "**MySQL 8.0 Reference Manual.**" Available: <https://dev.mysql.com/doc/>. Accessed: July 2026.
- A. Silberschatz, H. F. Korth, and S. Sudarshan, *Database System Concepts*, 7th ed. McGraw-Hill Education, 2020.